Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations

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Abstract. Multimodal Emotion Recognition in Conversations (MERC) aims to recognize human emotions for each utterance via visual, audio, and linguistic modalities. Its potential uses in affective computing and human-computer interaction have drawn a lot of attention in recent years. Among the various approaches proposed to solve this problem, graph based approaches have emerged as some of the most promising ones. However, because this pixel level feature extraction is noisy , the visual modality frequently makes only a small contribution or is underutilized in many graph based MERC frameworks. Traditional visual encoders generally focus on geometric and textual patterns to produce feature maps or patch representations. These representations often fail to sufficiently express the facial and behavioral cues leading to weak cross modal alignment with the other semantically rich modalities. To address this problem, we propose a Vision Language Model (VLM) based feature extraction approach that captures facial expressions and behavioral cues from the visual modality and encodes them as semantically meaningful representations. Using this approach, the raw video for each utterance is translated into a descriptive and meaningful textual representation through a Vision Language Model (VLM) with structured prompts designed to capture facial expressions, behavioral cues, and body language. The resulting semantic visual features are then incorporated into graph based MERC frameworks as a replacement for low level visual features, reducing noise in the visual modality while increasing the model's reliance on meaningful visual information through the VLM. This proposed method is expected to solve this problem of underutilization of visual modality while increasing the overall accuracy of MERC.

Keywords: First keyword · Second keyword · Third keyword

1 Introduction

The proper interpretation and communication of emotions are crucial, as they significantly influence interactions, conversations, and decision making processes. As this modern world advances to an AI dependent future it has become essential to ensure that these emerging AI systems and computers are capable of

recognizing emotions at a level comparable to, or exceeding, human ability. But so far understanding human emotions in conversations has been a fundamental challenge in human–computer interaction and affective computing [5]. Multimodal Emotion Recognition in Conversations (MERC) aims to identify the emotion expressed in each utterance by jointly leveraging information from multiple modalities, typically textual, acoustic, and visual signals [27, 35]. Unlike traditional single utterance emotion recognition, MERC must consider contextual dependencies across speakers and time, making it a more complex and realistic setting [10, 23].Over the past years, numerous approaches have been adopted to solve this problem, starting from early fusion based techniques and transformer driven multimodal interaction models to more structured and context aware graph based architectures [10,14,31,33,34]. Recent advances have demonstrated that the graph based architectures, particularly those inspired by Graph Convolutional Networks (GCNs) [17], are highly effective for MERC [10,14]. By modeling conversational structure as a graph, where utterances are treated as nodes and contextual relationships as edges, these approaches capture both inter speaker and intra speaker dependencies [10]. As a result, graph based MERC models achieve competitive performance on benchmark datasets and are considered among the most promising approaches for conversational emotion recognition [14].

Despite these advances these graph based approaches still face significant problems.One of them is that the visual modality often remains underutilized in many graph based MERC frameworks [14,19]. Traditional visual encoders, typically based on convolutional neural networks (CNNs) [13] or vision transformers (ViTs) [7], extract low level geometric, texture based, or patch level representations from raw video frames. While effective for general visual recognition tasks, these representations often fail to capture subtle behavioral cues and body language that are crucial for emotion understanding [20]. Consequently, the visual modality may contribute less than textual or acoustic modalities, which naturally encode higher level semantic information [11].

Despite advancements, graph based Multimodal Emotion Recognition in Conversation (MERC) approaches still face significant limitations , one of them being underutilization of a specific modality. Most of the currently existing graph based MERC frameworks are prone to this limitation underutilizing visual modality while giving priority to the textual modality [14, 19].

Traditional visual encoders relied upon in these baselines, such as Convolutional Neural Networks (CNNs) [13] or Vision Transformers (ViTs) [7], are fundamentally designed to extract abstract spatial features, structural hierarchies, or patch level representations from isolated, static video frames. While adequate for general object recognition, these representations create a semantic gap, failing to explicitly capture the subtle behavioral cues, interaction with the other characters in the scene and dynamic body language necessary for complex emotion understanding [20]. Moreover, in standard MERC pipelines, the temporal evolution of emotion within an utterance is largely disregarded. Because graph networks require a fixed size node representation for structural

modeling [10, 14], frame level visual embeddings are typically aggregated into utterance level representations via statistical pooling operations, as adopted in multimodal benchmarks [35]. This aggregation effectively collapses the intra utterance temporal dimension. As a result, dynamic behavioral shifts (such as a speaker transitioning from a frustrated frown to a resigned smile) are mathematically averaged into a single static and potentially emotionally ambiguous feature vector [33].

Furthermore, real world conversational videos are prone to noise, including illumination variations, occlusions, and background clutter. This noise severely degrades the quality of pixel level features. Consequently, the visual modality tends to contribute less to emotion understanding compared to textual or acoustic modalities, which inherently encode higher level semantic information.

This modality imbalance leads to weak cross modal alignment in multimodal fusion. Textual features derived from pretrained language models encode high level semantics [6], and acoustic features capture prosodic patterns associated with affect [30]. In contrast, low level visual embeddings remain largely appearance based, limiting effective multimodal integration, particularly in graph based architectures where node representations critically influence relational reasoning [10].

To address this limitation, we propose rethinking visual feature extraction for graph based MERC by moving from abstract spatial embeddings to contextualized, semantically meaningful visual descriptions. Specifically, we leverage advanced Video Language Models, specifically VidEmo-3B (built upon the Qwen2.5-VL-3B architecture [1]), which are highly capable of interpreting dynamic visual content and generating temporally grounded semantic representations. Instead of directly feeding mean-pooled, emotionally ambiguous visual vectors into the graph network, we translate each utterance level video segment into a temporally structured textual narrative. This narrative explicitly tracks the evolution of facial expressions, head movements, and body language from the beginning to the end of the utterance, preserving the critical intra utterance dynamics that traditional pipelines can not retain.

Through carefully designed prompts, the VLM extracts affect relevant visual information and encodes it into descriptive semantic representations. These generated textual features serve as high level visual summaries that are better aligned with linguistic and acoustic modalities [28]. The resulting semantic visual features are then integrated into graph based MERC architectures, replacing traditional low level visual embeddings. By doing so, we reduce visual noise, improve cross modal consistency, and encourage the model to rely on meaningful visual signals rather than appearance based artifacts.

In summary, this work makes the following contributions:

- We identify and analyze the underutilization of the visual modality in graph based MERC frameworks due to noisy and low level feature representations.
- We propose a VLM based semantic visual feature extraction approach that transforms raw video into structured textual descriptions capturing emotion relevant cues.

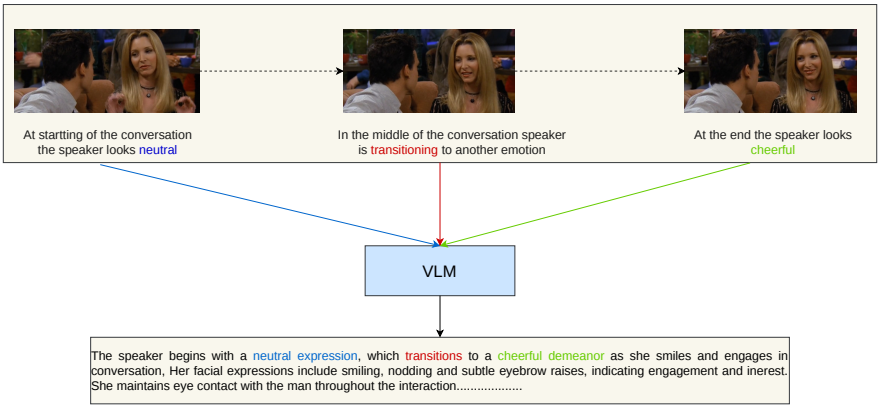

Fig. 1: Overview of the proposed semantic visual preprocessing pipeline.

– We integrate these semantic visual features into graph based MERC architectures, promoting stronger cross modal alignment.

By bridging the gap between visual perception and semantic understanding, our approach offers a new direction for enhancing the role of visual information in multimodal conversational emotion recognition.

2 Related Work

2.1 Emotion Recognition in Conversations and Multimodal Extensions

Emotion Recognition in Conversations (ERC) seeks to identify the emotional state expressed in each utterance by considering the context of preceding dialogue turns and speaker interactions [31]. Early sequence based methods relied on recurrent architectures such as RNNs and GRUs to capture temporal dependencies and track speaker states throughout a dialogue [11, 23]. While effective for short range context modeling, these approaches often struggle with long range dependencies and complex inter speaker influences.

Multimodal Emotion Recognition in Conversations (MERC) extends ERC by incorporating multiple modalities (typically text, acoustic, and visual) to better reflect how humans convey emotions [27, 35]. Textual information captures lexical and semantic content, acoustic signals encode prosodic and paralinguistic cues [30], and visual data contributes facial expressions and body language [20]. MERC's multimodal nature poses additional challenges, including modality alignment, noise handling (especially in the visual and acoustic channels), and effective cross modal fusion [31].

2.2 Approaches to MERC

To address these challenges, several modeling strategies have been proposed:

Sequence and Context Based Methods. Sequence models prioritize temporal dependencies and contextual flow. Early works such as DialogueRNN [23] process dialog turns sequentially, learning conversational context and speaker interactions through recurrent structures. While these approaches improve performance over context agnostic classifiers, they lack explicit mechanisms for relational reasoning between speakers or across modalities.

Fusion Based Methods. Fusion approaches aim to integrate multimodal information through strategies such as early concatenation, late decision fusion, cross modal attention, and transformer based fusion blocks.

Early / Late Fusion: Early fusion concatenates modality features at the input level, while late fusion combines modality specific predictions at the decision stage. These strategies are widely used baselines in multimodal sentiment and emotion analysis [24–26, 36].

Attention Based Fusion: Attention mechanisms dynamically weight modality contributions. The Multimodal Transformer [33] introduces cross modal attention for aligning text, audio, and visual streams, while MISA [12] learns modality invariant and modality specific representations through attention based fusion.

Transformer Based Fusion: Transformer architectures jointly model intra and inter modality dependencies using self and cross attention, enabling richer multimodal interaction [33].

Emotion Shift Aware Fusion: DialogueRNN [23] models emotion dynamics and speaker state tracking, and later multimodal variants incorporate shift aware mechanisms to better capture emotional transitions in dialogue [11].

While fusion strategies enhance cross modal interaction, they typically do not explicitly capture conversational structure or speaker relationships, which are crucial for nuanced emotional understanding in dialogue.

2.3 Graph Based MERC Approaches

Graph based approaches have emerged as one of the most powerful and widely adopted strategies for MERC because they directly encode conversational structure and relational dependencies into the learning process [14,19]. In this paradigm, utterances are represented as nodes, and edges encode speaker interactions, temporal continuity, and cross modal links [17], enabling structured graph propagation across the dialogue.

DialogueGCN [10] first formalized ERC as graph reasoning by modeling both intra speaker and inter speaker relations using graph convolution, showing superior performance over sequential models. This method laid the groundwork

for structural reasoning in conversational emotion tasks, highlighting the importance of relational context.

Subsequent graph extensions to MERC expanded the graph structure to include multimodal interactions. For example, MMGCN [14] and its variants build modality aware graphs to learn cross modal relationships jointly, and hypergraph based models capture higher order interactions among modalities and speakers.

A more recent line of work incorporates auxiliary objectives and enhanced graph learning. **Joyful** [18] integrates modality fusion with contrastive learning within a graph framework, constructing multimodal interaction graphs and employing contrastive objectives to encourage discriminative representations across modalities, addressing traditional issues such as over smoothing in graph embeddings and improving cross modal alignment.

GraphSmile [19] represents a state of the art graph centric MERC architecture introducing two key components: *Graph Structure Fusion (GSF)* [19], which constructs bipartite dialogue graphs [14] between every pair of modalities and alternates information propagation across these graphs to prevent any single structural path from dominating; and *Sentiment Dynamics Perception (SDP)* [19], an auxiliary task that explicitly predicts whether sentiment shifts occur between consecutive utterances to guide the model toward emotional transitions.

Despite its strong performance, GraphSmile, like most existing graph based MERC methods depends on conventional visual encoders (*e.g.*, CNN backbones [13] or handcrafted facial features [2]) that produce static, low level visual embeddings. These embeddings are typically mean-pooled over frames, collapsing intra utterance temporal dynamics and limiting the visual modality's contribution to emotional reasoning. Graph based approaches excel at structurally modeling dialogues and multimodal interactions, but their effectiveness is often constrained by the quality of unimodal representations feeding into the graph.

2.4 Vision Language and LLM Based Approaches for Emotion Recognition

To overcome limitations of low level visual features, recent research has explored Vision Language Models (VLMs) and multimodal Large Language Models (LLMs) for emotion recognition tasks. VLMs pretrained on large image/video text datasets [28] produce semantically meaningful representations of visual content: they generate textual or semantic descriptions from visual inputs, capturing behaviors, expressions, and dynamics that are difficult for CNN [13] or ViT [7] features to represent, and these descriptions are inherently aligned with language modalities [6], facilitating cross modal fusion and multimodal reasoning.

Recent works demonstrate that VLMs can capture emotionally relevant visual cues. Models such as VidEmo-3B [37] leverage video–language pretraining to describe visual affective cues over time, producing semantically grounded visual narratives. Task specific adaptations like Emotion-Qwen [15] build hybrid frameworks that integrate emotion reasoning into general vision–language understanding by combining specialized emotion experts with general VL backbones.

Evaluation studies of general VLMs show that, with appropriate prompting or fine tuning, vision–language models can recognize nuanced emotional expressions and behaviors that traditional encoders miss.

Multimodal LLM integrations extend this idea further. Works like BeMERC [9] propose behavior aware MLLM frameworks for MERC, using multimodal LLMs tuned to behavior features (*e.g.*, facial motion, body language) to directly predict conversational emotions, showing that LLMs can leverage both visual and linguistic context effectively. Prompt based or zero shot methods demonstrate that general large vision language systems such as CLIP [28] or Qwen2.5-VL [1] can be prompted to interpret emotional cues in video or images without explicit retraining, offering flexible emotion inference.

These VLM/LLM approaches address two key limitations in traditional MERC: visual modality underutilization, by generating high level semantic visual descriptions rather than low level features [20]; and cross modal alignment challenges, by inherently aligning visual and language modalities through shared semantic outputs [28]. Integrating semantic VLM representations into structured MERC frameworks, especially graph based methods, emerges as an important frontier for multimodal emotion recognition.

2.5 Positioning of Our Work

Our work builds upon the strengths of graph based MERC, particularly GraphSmile [19], while addressing the underutilization of visual information. Instead of using static, low level visual embeddings [2, 13], we introduce a VLM based semantic visual preprocessing stage [1, 37] that converts raw video segments into temporally structured textual narratives capturing facial expressions, gaze shifts, and body language evolution within an utterance.

These semantic visual descriptions are encoded into dense vectors [6] and integrated into the Graph Structure Fusion (GSF) pipeline [19]. This design preserves intra utterance temporal dynamics, improves cross modal alignment, reduces reliance on noisy visual appearance features, and enhances the model's ability to leverage visual cues in relational reasoning across conversation turns. By combining high level semantic visual reasoning from VLMs with structured graph propagation, our approach advances MERC beyond existing graph based and VLM based methods.

3 Methodology

The proposed framework builds upon the GraphSmile architecture [19] by introducing a novel approach that converts raw visual inputs into meaningful semantic representations before multimodal fusion.

While prior graph based Multimodal Emotion Recognition in Conversation (MERC) systems relied on conventional visual encoders (such as DenseNet for extracting abstract spatial features [16] and OpenFace for tracking deterministic facial mechanics like Action Units [2]), our methodology moves beyond these

restricted appearance descriptors. By leveraging a temporally aware Vision Language Model (VLM) [1,28], we encode the rich, semantically grounded behavioral signals present in human speech.

The overall architecture comprises three main stages:

1. The independent extraction of utterance level features across textual, acoustic, and visual modalities.
2. The construction of multimodal dialogue graphs for alternating propagation via the Graph Structure Fusion (GSF) module [19].
3. Multitask classification incorporating a Sentiment Dynamics Perception (SDP) module [19].

3.1 Utterance Level Feature Extraction

Each modality is processed independently at the utterance level before being integrated into the multimodal graph. In contrast to the visual channel, the textual and acoustic modality remains structurally unchanged relative to the GraphSmile baseline [19], preserving a controlled point of comparison.

Textual Feature Extraction. For the textual modality, utterance transcripts are encoded uniformly across both datasets using a pretrained transformer language model, specifically RoBERTa [21]. The `<s>` (classification) token representation of each transcript forms the textual feature vector, X^t, capturing lexical and syntactic sentiment signals.

Acoustic Feature Extraction. The acoustic features, X^a, capture paralinguistic speech characteristics including fundamental frequency (F0), energy, spectral properties, and temporal statistical functionals that reflect prosody and pitch dynamics. For both the IEMOCAP (6-way) [4] and MELD [27] datasets, these utterance level features are extracted using the openSMILE toolkit with the IS10 configuration [8].

Semantic Visual Feature Extraction. Traditional visual encoders often fail to fully harness conversational cues because they are restricted to extracting context blind spatial features or isolated facial mechanics, creating a semantic gap between raw physical movements and complex emotional states. The core methodological contribution addresses this by redefining the visual feature extraction process through a three stage semantic preprocessing pipeline.

Stage 1: Dataset Specific Video Preprocessing. To ensure the VLM accurately analyzes the correct subject, we apply tailored preprocessing strategies based on the native format of each dataset. The IEMOCAP dataset [4], which consists of continuous, multi speaker dialogue recordings, undergoes two explicit preprocessing steps. First, the sessions are temporally segmented into discrete, utterance level clips using provided timestamp annotations; second, each frame is

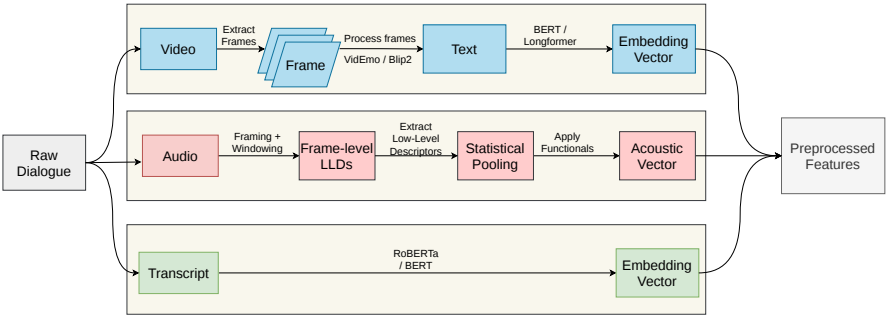

Fig. 2: The multimodal preprocessing pipeline for extracting utterance-level textual, acoustic, and visual features.

spatially cropped to physically isolate the active speaker and eliminate visual interference. In contrast, the MELD dataset [27] inherently provides presegmented utterance level clips, bypassing the need for physical cropping. Instead, spatial isolation for MELD is achieved entirely through zero shot prompting, wherein the VLM is explicitly instructed to autonomously identify and focus exclusively on the active speaker within the uncropped, multi person frame.

Stage 2: VLM Temporal Analysis. Each utterance clip is processed by VidEmo-3B [37] (built on the Qwen2.5-VL-3B backbone [1]). The model is driven by a highly structured prompt instructing it to focus exclusively on the active speaker and ignore any other visible participants. The VLM maps the speaker's emotional evolution by dividing the clip into a temporal progression (beginning, middle, and end). For each segment, the model grounds its analysis in specific visible signals across four dimensions:

- **Facial Expressions:** Specific muscle movements of the brow, lips, eyes, jaw, and cheeks.
- **Head & Gaze:** Tilts, nods, shakes, and eye direction.
- **Body Language:** Posture shifts, gestures, and physical tension or relaxation.
- **Instantaneous Emotion:** The most likely emotional state, explicitly supported by the aforementioned visual cues.

The temporal analysis concludes with a one sentence summary of the overall emotional arc (*e.g.*, *starts neutral* $\rightarrow$ *builds frustration* $\rightarrow$ *brief smile at end*). By enforcing that every observation is grounded in a specific visible signal from the speaker, this stage generates a semantically dense, temporally aware description of affective behavior, bypassing the noise of traditional frame averaging encoders.

Stage 3: Text Encoding of VLM Output. To bridge the symbolic output of the VLM with the continuous feature space expected by the graph network, the free text description is encoded using a pretrained BERT model [6]. The resulting fixed dimensional dense vector serves as the new visual modality feature, X^v.

Encoding visual behavior in the same semantic space as the transcript fosters superior cross modal alignment within the GSF module [19].

3.2 Graph Structure Fusion (GSF) Module

The Graph Structure Fusion (GSF) module processes dialogue graphs layer by layer, alternating between sequential edges for temporal flow and speaker edges for personal consistency. This back-and-forth approach prevents any single relationship from dominating and effectively separates a speaker's internal emotional shifts from the conversation's external influences.

Multimodal Dialogue Graphs To understand how utterances relate to each other across different sensory inputs, we build three bipartite dialogue graphs [14, 19]. Each graph connects two specific modalities. For example, the textual visual graph is defined as $\mathcal{G}^{tv} = (\mathcal{N}^t, \mathcal{N}^v, \mathcal{E}^{t\leftarrow v}, \mathcal{E}^{v\leftarrow t})$. Here, $\mathcal{N}^t$ and $\mathcal{N}^v$ stand for the sets of textual and visual nodes. The edge sets $\mathcal{E}^{t\leftarrow v}$ and $\mathcal{E}^{v\leftarrow t}$ represent the connections directed from visual to textual nodes, and vice versa.

- **Nodes:** Every utterance u_i is represented by a textual node n_i^t and a visual node n_i^v.
- **Edges:** We only draw edges between nodes of opposite modalities. To keep the graph from becoming too dense, we use a sliding window approach. A textual node n_i^t connects to visual nodes from the past P steps (i.e., $n_{i-P}^v, \ldots, n_{i-1}^v$) and the future F steps (i.e., $n_{i+1}^v, \ldots, n_{i+F}^v$).
- **Edge Weights:** These connections are directed to show the specific flow of information. We assign trainable weights to these edges, which start at a value of 1 and are fine tuned as the model trains.
- **Adjacency Matrix:** All of these node and edge relationships are stored in an adjacency matrix. This matrix is the core structural input for the graph neural network.

Alternate Graph Propagation Graph Neural Networks (GNNs) [17, 29] are highly effective at uncovering both cross modal and intra modal relationships. In our design, each dialogue graph passes information through multiple simplified convolutional layers. It alternates between extracting inter modal and intra modal features layer by layer, calculated as:

$$X^{(l)} = AX^{(l-1)}\Theta^{(l)}, \quad l = 1, 2, \ldots, L$$

Here, A is the dialogue graph's adjacency matrix, $X^{(l)}$ is the output feature at layer l, and $\Theta^{(l)}$ is the trainable weight matrix for that layer. For the textual-visual graph, the starting input $X^{tv,(0)}$ simply combines the raw textual features $[x_1^t, \ldots, x_M^t]$ and visual features $[x_1^v, \ldots, x_M^v]$.

Next, we use residual connections [13] to sum the feature representations from every layer:

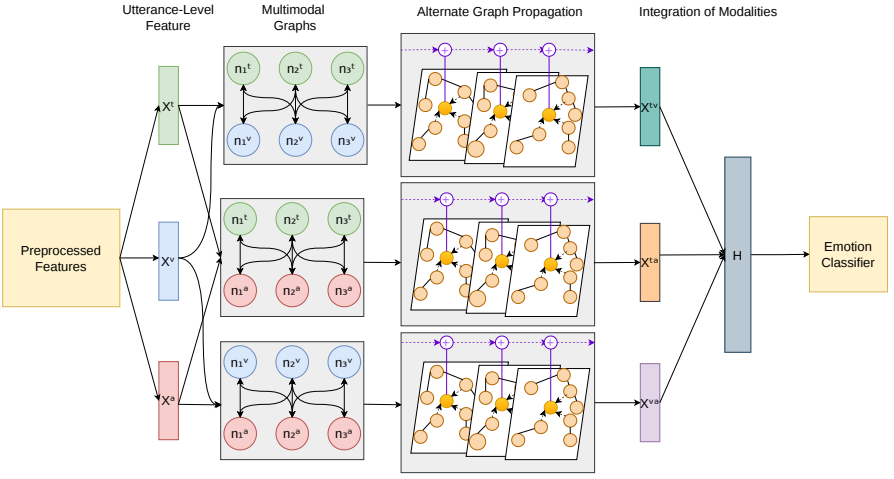

Fig. 3: The GraphSmile [19] architecture: bipartite multimodal dialogue graphs connecting utterance nodes across modality pairs, propagated through the Graph Structure Fusion (GSF) module.

$$X = X^{(0)} + \mathrm{FC}^{(1)}(X^{(1)}) + \cdots + \mathrm{FC}^{(L)}(X^{(L)})$$

The fully connected layer $\mathrm{FC}(\cdot)$ includes a linear transformation, a LeakyReLU [22] activation, and dropout [32]:

$$\mathrm{FC}(X) = \mathrm{Dropout}(\mathrm{LeakyReLU}(X\Theta_f + b_f))$$

This setup serves two purposes: it lets the network dynamically choose which layer's information is most useful, and it boosts the overall expressive power of the GSF module.

Multimodal Integration After processing the textual-visual ($\mathcal{G}^{tv}$), textual-acoustic ($\mathcal{G}^{ta}$), and visual-acoustic ($\mathcal{G}^{va}$) graphs through their own separate GSF modules, we get three distinct outputs: X^{tv}, X^{ta}, and X^{va}. We add these together to create a final, unified feature representation:

$$\begin{aligned} H = \sigma\big((X^{t\leftarrow v})\Theta_h\big) + \sigma\big((X^{v\leftarrow t})\Theta_h\big) \\ + \sigma\big((X^{t\leftarrow a})\Theta_h\big) + \sigma\big((X^{a\leftarrow t})\Theta_h\big) \\ + \sigma((X^{v\leftarrow a})\Theta_h) + \sigma((X^{a\leftarrow v})\Theta_h) \end{aligned} \tag{1}$$

In this equation, σ is the activation function (like LeakyReLU) and Θ_h is a shared learning parameter. This combined matrix H effectively captures all the complex, multi sensory emotional signals in the conversation.

3.3 Emotion Classification

After the graph propagation and integration phase, the resulting feature matrix H is passed into a multitask classification head. For the primary task of emotion classification, H goes through a linear classifier and a softmax function to predict the exact emotional category. This is formulated as:

$$\hat{Y} = \text{softmax}(H\Theta_y + b_y)$$
$$\hat{e}_i = \text{argmax}(\hat{y}_i[\tau])$$

Here, Θ_y and b_y are the learned weights and biases. The vector $\hat{y}_i$ holds the emotion probabilities for utterance u_i, and $\hat{e}_i$ is the final predicted emotion label. We optimize the network using categorical cross-entropy loss:

$$\mathcal{L}_e = -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{C_e} y_{i,j}\log\hat{y}_{i,j}$$

where M is the total number of utterances, C_e is the number of emotion classes, $y_{i,j}$ is the true label, and $\hat{y}_{i,j}$ is our model's prediction.

3.4 Sentiment Dynamics Perception (SDP) Module

Existing dialogue models often focus on static context, missing crucial inter utterance sentiment changes. To address this, we introduce the Sentiment Dynamics Perception (SDP) module [19] as a vital auxiliary task. The SDP module frames conversation dynamics as a pairwise prediction problem, tracking whether sentiment shifts or remains constant between consecutive utterances. By jointly optimizing this auxiliary shift loss with the main emotion classification loss, the network is strictly guided to learn the dialogue's emotional momentum. This joint supervision improves class separation (pulling similar sentiments closer and pushing differing ones apart), while perfectly aligning with the temporal descriptions created during the VLM preprocessing stage.

Shift Label We define a sentiment shift label $o_{(i,j)}$ for any pair of utterances u_i and u_j. If the sentiment changes between the two utterances, the label is 1; if the sentiment remains the same, the label is 0:

$$o_{(i,j)} = \begin{cases} 1, & \text{if } s_i \neq s_j, \\ 0, & \text{if } s_i = s_j. \end{cases}$$

where s_i denotes the ground truth sentiment state of utterance u_i.

Shift Classification We construct pairwise shift features T from the fused representations and pass them through a softmax classifier to predict the shift probability $\hat{Z}$:

$$\hat{Z} = \text{softmax}(T\Theta_z + b_z)$$

where Θ_z and b_z are learnable parameters. The module is optimized using a standard cross-entropy loss function, denoted as $\mathcal{L}_o$.

3.5 Sentiment Classification

To extend our model to the Multimodal Sentiment Analysis in Conversation (MSAC) task, we add a sentiment classifier. The fused feature matrix H is processed through a linear layer and a softmax activation:

$$\hat{W} = \text{softmax}(H\Theta_w + b_w)$$

Here, Θ_w and b_w are learnable parameters, and $\hat{W}$ contains the predicted sentiment probabilities. This task is supervised by a sentiment classification cross-entropy loss, denoted as $\mathcal{L}_s$.

3.6 Total Training Objective

To simultaneously extract both emotional and sentimental cues, the primary emotion recognition task is jointly learned alongside the sentiment and shift classification tasks. The overall network parameters are optimized using a combined loss function:

$$\mathcal{L}_{total} = \mathcal{L}_e + \lambda_s \mathcal{L}_s + \lambda_o \mathcal{L}_o + \beta ||\Theta_{total}||$$

where $\mathcal{L}_e$ is the primary emotion classification loss, λ_s and λ_o are trade-off hyperparameters balancing the auxiliary tasks, β is the L_2 regularization weight, and Θ_{total} represents all trainable parameters in the model.

4 Experimental Details

4.1 Datasets Overview

IEMOCAP [4]**.** The dataset consists of continuous, dyadic conversational sessions recorded with a dual-camera setup, presenting both actors side-by-side in a single unsegmented video frame. The accompanying audio is recorded in stereo, capturing each speaker on a dedicated independent microphone channel.

MELD [27]**.** This dataset is constructed from episodes of the television series *Friends*, representing a multi-party conversational environment. The video data is inherently provided as pre-segmented, utterance-level clips.

Table 1: Dataset statistics for IEMOCAP and MELD.

Dataset	Dialogues	Utterances	Classes	Train / Val / Test
IEMOCAP	151	7,433	6	5,810 / —/ 1,623*
MELD	1,432	13,708	7	9,989 / 1,108 / 2,611

4.2 Utterance-Level Video Preparation

IEMOCAP. Since IEMOCAP provides continuous dual-speaker recordings, the following preparation steps are applied:

- Each conversation video is segmented into utterance-level clips using official timestamps.
- Spatial cropping is applied to isolate the active speaker from the side-by-side frame.
- The stereo audio stream is separated so that only the corresponding speaker's channel is retained.
- The output is a speaker-focused, utterance-level video clip.

This ensures that the visual modality represents only the speaking individual at each dialogue turn.

MELD. MELD already provides pre-segmented utterance-level clips:

- No spatial cropping is required.
- Each clip corresponds directly to a single speaker utterance.
- Speaker focus is handled during prompting rather than preprocessing.

4.3 Semantic Visual Feature Extraction

Video-to-Text Generation. Vision Language Models (VLMs) are utilized to generate descriptive text for each utterance. The VidEmo-3B model [37], built upon the Qwen2.5-VL backbone [1], is applied to both datasets.

Frame Sampling. IEMOCAP videos are sampled at a fixed rate of 1 frame per second (FPS). MELD videos use an adaptive strategy of 1 FPS for clips under 15 seconds, capped at a maximum of 16 frames for longer clips.

Prompt Structure. The prompt requests a temporal progression (beginning, middle, and end) detailing facial expressions, head and gaze, body language, and instantaneous emotion, concluding with a one-sentence emotional arc summary. For MELD, zero-shot instructions are included to focus exclusively on the active speaker.

Batching and Computation. For IEMOCAP, inference was executed with a batch size of 4, requiring over 7 hours. For MELD, inference was executed with a batch size of 2, requiring over 10 hours.

Text-to-Vector Encoding. The generated descriptions are encoded using two models evaluated in separate experimental conditions:

- **Longformer** [3] (`allenai/longformer-base-4096`): inputs are processed without truncation; a 768-dimensional embedding is extracted from the `[CLS]` token.
- **BERT** [6] (`bert-base-uncased`): inputs are truncated to a maximum of 512 tokens; a 768-dimensional embedding is extracted from the `[CLS]` token.

4.4 Implementation Details

Hardware and Environment. All experiments are conducted on an NVIDIA GeForce RTX 3060 (12 GB VRAM) using a Conda environment with PyTorch 2.8.0+cu126.

Table 2: Hyperparameter settings for IEMOCAP and MELD.

Hyperparameter	IEMOCAP	MELD
Epochs	120	50
Batch Size	16	16
Learning Rate	1×10^{-4}	7×10^{-5}
Hidden Dimension	512	384
Sliding Window	17	3
Shift Window	19	3
Heterogeneous Graph Layers	7	4
Dropout Rate	0.2	0.2
Loss Weights	1.0, 1.0, 0.7	0.0, 0.5, 0.2

4.5 Integration into Graph Based MERC Framework

Feature Substitution and Dimensionality. The 768-dimensional visual embeddings generated via the VLM and subsequent text encoding are substituted for the original 342-dimensional handcrafted visual descriptors in the dataset manifest. A linear projection layer is implemented at the input of the visual processing stream to map the 768-dimensional vectors into the hidden dimension space of the graph architecture. Textual and acoustic features remain identical to the GraphSmile baseline [19], with dimensions of 1024 (RoBERTa) and 1582 (openSMILE IS10) for IEMOCAP, and 1024 and 300 respectively for MELD.

Evaluation Metrics. Accuracy (ACC) and Weighted F1-score (WF1) serve as the primary evaluation metrics for overall model performance. The F1-score is further utilized to evaluate classification performance for each individual emotion category.

Training and Optimization Strategy. The model follows the multi-task optimization strategy of GraphSmile [19], jointly learning the primary emotion recognition task alongside sentiment and sentiment-shift classification. All structural components of the Graph Structure Fusion (GSF) module and the Sentiment Dynamics Perception (SDP) module are maintained as per the original implementation to ensure a controlled comparison of visual feature effectiveness.

5 Conclusion

The paper ends with a conclusion.

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Please insert your acknowledgments here.

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